

Nonlinear Spectroscopy I: Time-dependent Perturbation Theory and Nonlinear Response Theory

Kai Cui

1 Perturbation and Response Theories in Linear Spectroscopy

In linear spectroscopy, the optical processes can be described using both first-order time-dependent perturbation theory (a.k.a. Fermi's golden rule) and linear response theory. Using the absorption as an example, Fermi's golden rule describes the transition rate of a molecule from its ground state to excited state under the perturbation of the electric field of the incident light, $\mathbf{E}(t) = \mathbf{E}_0 \cos(\omega t)$, as

$$k = \frac{2\pi}{\hbar} \left| \frac{1}{2} \boldsymbol{\mu}_{ge} \cdot \mathbf{E}_0 \right|^2 \delta(E_g - E_e + \hbar\omega) \quad (1)$$

where $\boldsymbol{\mu}_{ge}$ is the transition dipole moment, $\mathbf{E}_0$ is the amplitude of the external electric field, ω is the frequency of the incident light. Due to the delta function, the transition rate is nonzero only when ω equals to the transition frequency, $\omega_{ge} = \frac{E_e - E_g}{\hbar}$. The absorbance is proportional to the transition rate, which in turn is proportional to the norm square of the transition dipole moment.

In linear response theory, the Fourier transformed linear response function (polarizability tensor) is

$$\alpha_{\xi\eta}(\omega) = -\frac{1}{\hbar} \sum_c \left[\frac{(\mu_\xi)_{ac}(\mu_\eta)_{ca}}{\omega - \omega_{ac} + i\varepsilon} - \frac{(\mu_\eta)_{ac}(\mu_\xi)_{ca}}{\omega + \omega_{ac} + i\varepsilon} \right] \quad (2)$$

where ξ and η labels the Cartesian components of a vector and a tensor, ω_{ac} is the transition frequency between states a and c , $\omega_{ac} = \frac{E_c - E_a}{\hbar}$. The average polarizability of the molecule is

$$\alpha(\omega) \equiv \frac{1}{3} \text{Tr} \alpha_{\xi\eta}(\omega) = -\frac{2}{3\hbar} \sum_c \frac{\omega_{ac} |\boldsymbol{\mu}_{ac}|^2}{(\omega + i\varepsilon)^2 - \omega_{ac}^2} \quad (3)$$

The linear polarizability peaks at the transition frequency, and the peak height is proportional to the norm square of the transition dipole moment. **For absorption, calculating the quantum mechanical transition rate and calculating the Fourier transformed response function give the same spectra.**

For the case of light scattering (use Rayleigh scattering as an example), from a classical perspective, the electric field of the light drives an oscillation of the molecular dipole moment. This oscillation thus has a frequency ω same as the frequency of the incident light. In linear response theory, the induced dipole moment in frequency domain is

$$\boldsymbol{\mu}_{\text{ind}}(\omega) = \boldsymbol{\alpha}(\omega) \delta(\omega) \cdot \mathbf{E}_0 \quad (4)$$

This oscillating induced dipole will radiate electromagnetic wave with the same frequency, which is the scattered light. In classical electrodynamics, the field strength of the dipole radiation is proportional to the second time derivative of the dipole moment, $\ddot{\boldsymbol{\mu}}(t)$,

$$\ddot{\boldsymbol{\mu}}(t) = -\frac{1}{2\pi} \int d\omega \omega^2 \boldsymbol{\mu}_{\text{ind}}(\omega) e^{-i\omega t} = -\frac{1}{2\pi} \omega^2 \boldsymbol{\alpha}(\omega) \cdot \mathbf{E}_0 e^{-i\omega t} \quad (5)$$

Therefore, the intensity of the scattered light is proportional to the norm square polarizability.

From a quantum mechanical perspective, in the scattering process, the molecule absorb a photon and emit another photon. The rate constant of this transition can be calculated using second-order time-dependent perturbation theory. In the previous note on light scattering, we have shown that the second-order coupling is just the matrix element of the polarizability operator. If the molecule goes back to its initial state after the scattering, the outgoing photon must have the same frequency as the incoming photon, and the intensity of the scattered light ($\propto k$) is proportional to the norm square of the expectation value of the polarizability operator. **For Rayleigh scattering, second-order perturbation theory and nonlinear response theory also give the same result.**

Absorption, emission, and scattering can all be described using the linear response theory, they are therefore considered as linear spectroscopy. Nonlinear spectroscopy corresponds to the scenarios where the linear response theory fails. One simple example is the **multiwave mixing**, where the molecule interact with more than one radiation fields. In the literature and textbooks, the nonlinear spectroscopy are mostly modeled using nonlinear response theory, and the spectra are obtained from Fourier transformation or partial Fourier transformation of the nonlinear response functions. On the other hand, higher-order perturbation theory is seldom used. In this note, we will derive the transition rate constants of multiphoton processes using nonlinear time-dependent perturbation theory, and then derive the nonlinear response functions of the molecule. Using multiwave mixing as an example, we will show that the higher-order perturbation theory and nonlinear response theory still give the same result for nonlinear optical processes. Finally, we will comment on the advantages and limitations of the two approaches.

2 Time-dependent Perturbation Theory

Let $\hat{H}_0$ be the Hamiltonian of a free molecule. E_a and $|a\rangle$ are the eigenvalue and eigenstates of $\hat{H}_0$,

$$\hat{H}_0|a\rangle = E_a|a\rangle \quad (6)$$

The time evolution operator for the free molecule is given by

$$\hat{U}_0(t, t_0) = e^{-i(t-t_0)\frac{\hat{H}_0}{\hbar}} \equiv \hat{U}_0(t) \quad (7)$$

Using the dipole approximation, the molecule interact with a quantized electric field via

$$\hat{V} = -\hat{\boldsymbol{\mu}} \cdot \hat{\mathbf{E}} \quad (8)$$

where $\hat{\boldsymbol{\mu}}$ is the dipole operator. $\hat{\mathbf{E}}$ is the electric field operator, which is given by

$$\hat{\mathbf{E}}(\mathbf{r}) = i \sum_{\mathbf{k}, \lambda} \sqrt{\frac{\hbar \omega_{\mathbf{k}}}{2V n_r^2 \varepsilon_0}} \left(\mathbf{e}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} e^{i\mathbf{k} \cdot \mathbf{r}} - \mathbf{e}_{\mathbf{k}\lambda}^* \hat{a}_{\mathbf{k}\lambda}^\dagger e^{-i\mathbf{k} \cdot \mathbf{r}} \right) \quad (9)$$

Here $\hat{a}_{\mathbf{k}\lambda}^\dagger$ and $\hat{a}_{\mathbf{k}\lambda}$ are the creation and annihilation operators for the photon with wave vector $\mathbf{k}$ and polarization λ . $\mathbf{e}_{\mathbf{k}\lambda}$ is a unit vector that parallel to polarization direction. $\mathbf{e}_{\mathbf{k}\lambda}$ is perpendicular to $\mathbf{k}$. n_r is the refractive index of the medium.

The time evolution operator of this perturbed system is

$$\hat{U}(t, t_0) = \hat{U}_0(t, t_0) \hat{T} \exp \left\{ -\frac{i}{\hbar} \int_{t_0}^t d\tau \hat{U}_0^\dagger(\tau) \hat{V} \hat{U}_0(\tau) \right\} \quad (10)$$

where $\hat{T}$ is time ordering operator, if $\hat{O}(t)$ is a time-dependent operator, then

$$\hat{T}[\hat{O}(t_2) \hat{O}(t_1)] = \begin{cases} \hat{O}(t_1) \hat{O}(t_2) & \text{if } t_2 < t_1 \\ \hat{O}(t_2) \hat{O}(t_1) & \text{if } t_2 > t_1 \end{cases} \quad (11)$$

Expanding the exponential and apply the time ordering, we have

$$\hat{U}(t, t_0) = \hat{U}_0(t, t_0) + \sum_{n=1}^{\infty} \hat{U}^{(n)}(t, t_0) \quad (12)$$

where $\hat{U}^{(n)}(t, t_0)$ is the n -th order expansion of the time evolution operator,

$$\hat{U}^{(n)}(t, t_0) = \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \dots \int_{t_0}^{\tau_2} d\tau_1 \hat{U}_0(t, \tau_n) \hat{V} \hat{U}_0(\tau_n, \tau_{n-1}) \dots \hat{U}_0(\tau_2, \tau_1) \hat{V} \hat{U}_0(\tau_1, t_0) \quad (13)$$

The time variables in this equation is fully ordered, where $t \geq \tau_n \geq \tau_{n-1} \geq \dots \geq \tau_2 \geq \tau_1 \geq t_0$. Due to the time ordering, there is no $1/n!$ factor in Eq. (13) which should appear in normal Taylor expansion of an exponential function. The integrand can be understood in the following way. **In the n -th order expansion, the system interacts with the field n times. The system first propagates freely with the unperturbed Hamiltonian $\hat{H}_0$ from t_0 to τ_1 and interacts with the field at τ_1 ; then it propagates freely again with $\hat{H}_0$ from τ_1 to τ_2 and interacts with the field again; and so on. The n -fold integration run over all possible time sequences.**

We want to calculate the transition rate between molecular states due to the perturbation. Suppose that the molecule is at state $|a\rangle$ at t_0 . Let $|\psi(t)\rangle$ be its state at t , then the population of one final state $|b\rangle$ at time t is given by

$$P_b(t) = |\langle b | \psi(t) \rangle|^2 = |\langle b | \hat{U}(t, t_0) | a \rangle|^2 \quad (14)$$

The transition amplitude A_{ba} is defined as

$$A_{ba} = \langle b | \hat{U}(t, t_0) | a \rangle \quad (15)$$

Using the expansion Eq. (12), we have

$$A_{ba} = A_{ba}^{(0)} + A_{ba}^{(1)} + A_{ba}^{(2)} + A_{ba}^{(3)} + \dots \quad (16)$$

where

$$A_{ba}^{(0)} = e^{-i\frac{E_b(t-t_0)}{\hbar}} \delta_{ba} \quad (17a)$$

$$A_{ba}^{(1)} = -\frac{i}{\hbar} e^{-i(\phi_1+\phi_2)} V_{ba} \int_{t_0}^t d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} \quad (17b)$$

$$A_{ba}^{(2)} = -\frac{1}{\hbar^2} e^{-i(\phi_1+\phi_2)} \sum_c V_{bc} V_{ca} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 e^{i\tau_2\frac{E_b-E_c}{\hbar}} e^{i\tau_1\frac{E_c-E_a}{\hbar}} \quad (17c)$$

$$A_{ba}^{(3)} = \frac{i}{\hbar^3} e^{-i(\phi_1+\phi_2)} \sum_{c,d} V_{bc} V_{cd} V_{da} \int_{t_0}^t d\tau_3 \int_{t_0}^{\tau_3} d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 e^{i\tau_3\frac{E_b-E_c}{\hbar}} e^{i\tau_2\frac{E_c-E_d}{\hbar}} e^{i\tau_1\frac{E_d-E_a}{\hbar}} \quad (17d)$$

where $V_{ba} = \langle b|\hat{V}|a\rangle$ is the matrix element of the perturbation. $\phi_1 = \frac{E_b(t-t_0)}{\hbar}$ and $\phi_2 = \frac{E_b-E_a}{\hbar}t_0$ are phase factors.

The transition rate is defined via Pauli master equation,

$$\frac{d}{dt}P_b(t) = k_{a \rightarrow b}P_a(t) \quad (18)$$

where $P_a(t)$ is the norm square of the surviving amplitude, A_{aa} . If we only keep the first nonzero expansion term,

$$A_{aa} \approx A_{aa}^{(0)} = e^{-i\frac{E_a(t-t_0)}{\hbar}} \quad (19)$$

Thus $P_a(t) = |A_{aa}|^2 \approx 1$, and

$$\boxed{k_{a \rightarrow b} = \frac{d}{dt}|A_{ba}|^2} \quad (20)$$

Clearly, $A_{ba}^{(0)} = 0$ when $b \neq a$. The first-order term is

$$|A_{ba}^{(1)}|^2 = \frac{1}{\hbar^2} |V_{ba}|^2 \left| \int_{t_0}^t d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} \right|^2 \quad (21)$$

It's easy to show that

$$\begin{aligned} \frac{d}{dt} \left| \int_{t_0}^t d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} \right|^2 &= \frac{d}{dt} \left(\int_{t_0}^t d\tau e^{-i\tau\frac{E_b-E_a}{\hbar}} \times \int_{t_0}^t d\tau' e^{i\tau'\frac{E_b-E_a}{\hbar}} \right) \\ &= \int_{t_0}^t d\tau e^{i(t-\tau)\frac{E_b-E_a}{\hbar}} + \int_{t_0}^t d\tau e^{-i(t-\tau)\frac{E_b-E_a}{\hbar}} \\ &= \int_{-t+t_0}^{t-t_0} d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} \end{aligned} \quad (22)$$

The last line is derived by changing the variables $\tau_1 = t - \tau$ for the first integral in the second line and $\tau_2 = -(t - \tau)$ for the second one. If we assume that $t_0 \rightarrow -\infty$, then

$$\frac{d}{dt} \left| \int_{t_0}^t d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} \right|^2 = \int_{-\infty}^{\infty} d\tau e^{i\tau\frac{E_b-E_a}{\hbar}} = 2\pi\hbar\delta(E_b - E_a) \quad (23)$$

Therefore, **the first-order approximation of the transition rate is**

$$k_{a \rightarrow b}^{(1)} = \frac{2\pi}{\hbar} |V_{ba}|^2 \delta(E_b - E_a) \quad (24)$$

This is the Fermi's golden rule formula. When calculating V_{ba} for a spectroscopic process, the initial and final states are not just molecular states, but are direct products of a molecular state and a photon state. **Since $\hat{V}$ is a one-photon operator, the matrix element V_{ba} is nonzero only if the transition is a one-photon process, i.e. only one photon occupation number changes.** Such one-photon processes include absorption and emission. If more than one photon is involved in the transition, $V_{ba} = 0$, and higher-order terms must be considered.

The second-order term is

$$|A_{ba}^{(2)}|^2 = \frac{1}{\hbar^4} \left| \sum_c V_{bc} V_{ca} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 e^{i\tau_2 \frac{E_b - E_c}{\hbar}} e^{i\tau_1 \frac{E_c - E_a}{\hbar}} \right|^2 \quad (25)$$

The double time integral can be rewritten using the Heaviside step function,

$$\begin{aligned} & \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 e^{i\tau_2 \frac{E_b - E_c}{\hbar}} e^{i\tau_1 \frac{E_c - E_a}{\hbar}} \\ &= \int_{t_0}^t d\tau_2 \int_{t_0}^{+\infty} d\tau_1 \theta(\tau_2 - \tau_1) e^{i\tau_2 \frac{E_b - E_c}{\hbar}} e^{i\tau_1 \frac{E_c - E_a}{\hbar}} \\ &= \lim_{\varepsilon \rightarrow 0^+} \frac{i}{2\pi} \int_{t_0}^t d\tau_2 \int_{t_0}^{+\infty} d\tau_1 \int_{-\infty}^{+\infty} d\omega \frac{e^{-i\omega(\tau_2 - \tau_1)}}{\omega + i\varepsilon} e^{i\tau_2 \frac{E_b - E_c}{\hbar}} e^{i\tau_1 \frac{E_c - E_a}{\hbar}} \\ &= \lim_{\varepsilon \rightarrow 0^+} \frac{i}{2\pi} \int_{t_0}^t d\tau_2 \int_{-\infty}^{+\infty} d\omega \frac{1}{\omega + i\varepsilon} e^{-i(\omega - \frac{E_b - E_c}{\hbar})\tau_2} \int_{t_0}^{+\infty} d\tau_1 e^{i(\omega + \frac{E_c - E_a}{\hbar})\tau_1} \end{aligned} \quad (26)$$

In the third line, we used the integral representation of the step function,

$$\theta(t) = \lim_{\varepsilon \rightarrow 0^+} \frac{i}{2\pi} \int_{-\infty}^{\infty} d\omega \frac{e^{-i\omega t}}{\omega + i\varepsilon} \quad (27)$$

We first take $t_0 \rightarrow -\infty$ limit for the integral over τ_1 , which leads to a delta function, and then integrate over ω ,

$$\begin{aligned} & \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 e^{i\tau_2 \frac{E_b - E_c}{\hbar}} e^{i\tau_1 \frac{E_c - E_a}{\hbar}} \\ &= \lim_{\varepsilon \rightarrow 0^+} i \int_{t_0}^t d\tau_2 \int_{-\infty}^{+\infty} d\omega \frac{\delta(\omega + \frac{E_c - E_a}{\hbar})}{\omega + i\varepsilon} e^{-i(\omega - \frac{E_b - E_c}{\hbar})\tau_2} \\ &= \lim_{\varepsilon \rightarrow 0^+} \frac{i\hbar}{E_a - E_c + i\varepsilon} \int_{t_0}^t d\tau_2 e^{i\tau_2 \frac{E_b - E_a}{\hbar}} \end{aligned} \quad (28)$$

and

$$|A_{ba}^{(2)}|^2 = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\hbar^2} \left| \sum_c \frac{V_{bc} V_{ca}}{E_a - E_c + i\varepsilon} \right|^2 \left| \int_{t_0}^t d\tau e^{i\tau \frac{E_b - E_a}{\hbar}} \right|^2 \quad (29)$$

The time derivative of the last integral can be calculated in the same manner as in the first-order term, which leads to **the second-order approximation of the transition rate**,

$$k_{a \rightarrow b}^{(2)} = \frac{2\pi}{\hbar} \left| \sum_c \frac{V_{bc}V_{ca}}{E_a - E_c + i\varepsilon} \right|^2 \delta(E_b - E_a) \quad (30)$$

We have omitted the limitation sign in this equation for simplicity. Following the same procedure, it's easy to calculate the higher-order terms, for example, **the third-order approximation** is

$$k_{a \rightarrow b}^{(3)} = \frac{2\pi}{\hbar} \left| \sum_{c,d} \frac{V_{bc}V_{cd}V_{da}}{(E_a - E_c + i\varepsilon)(E_a - E_d + i\varepsilon)} \right|^2 \delta(E_b - E_a) \quad (31)$$

The first-order approximation describes a **direct transition** from state $|a\rangle$ to $|b\rangle$, while the higher order approximations describe an **indirect transition: the transition from state $|a\rangle$ to $|b\rangle$ must go through some intermediate state(s)**. When $V_{ba} = 0$, the direct transition is forbidden, but indirect paths could be allowed. Scattering of light is a two-photon process, the rate and cross section of scattering must be described using the second-order approximation. For transitions involve three photons, like three wave mixing, the third-order term is needed.

3 Nonlinear Response Theory

In this section we discuss the time evolution of molecular dipole moment under the perturbation of external field. Let $\hat{\rho}$ be the density operator of the system, then the expectation value of the dipole moment is

$$\mathbf{P}(t) = \text{Tr} \left(\hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) \hat{\rho}_0 \right) \equiv \left\langle \hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) \right\rangle \quad (32)$$

where $\hat{\rho}_0$ is the equilibrium density operator of the unperturbed system,

$$\hat{\rho}_0 = \frac{e^{-\beta \hat{H}_0}}{\text{Tr} e^{-\beta \hat{H}_0}} \quad (33)$$

$\langle \dots \rangle = \text{Tr}(\dots \hat{\rho}_0)$ denotes the ensemble average of over the unperturbed distribution. Substituting the perturbative expansion of the time evolution operator,

$$\hat{U}(t) = \hat{U}_0(t) + \lambda \hat{U}^{(1)}(t) + \lambda^2 \hat{U}^{(2)}(t) + \lambda^3 \hat{U}^{(3)}(t) + \dots \quad (34)$$

where the λ factor is used to collect the terms with the same order of perturbation, we have

$$\begin{aligned} \mathbf{P}(t) &= \left\langle \hat{U}_0^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}_0(t) \hat{\rho}_0 \right\rangle \\ &+ \lambda \left\langle \hat{U}_0^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}^{(1)}(t) + \hat{U}^{(1)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}_0(t) \right\rangle \\ &+ \lambda^2 \left\langle \hat{U}^{(1)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}^{(1)}(t) + \hat{U}_0^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}^{(2)}(t) + \hat{U}^{(2)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}_0(t) \right\rangle \\ &+ \lambda^3 \left\langle \hat{U}^{(1)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}^{(2)}(t) + \hat{U}^{(2)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}^{(1)}(t) + \hat{U}_0^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}^{(3)}(t) + \hat{U}^{(3)\dagger}(t) \hat{\boldsymbol{\mu}} \hat{U}_0(t) \right\rangle \\ &+ \dots \\ &= \mathbf{P}_0 + \lambda \mathbf{P}^{(1)}(t) + \lambda^2 \mathbf{P}^{(2)}(t) + \lambda^3 \mathbf{P}^{(3)}(t) + \dots \end{aligned} \quad (35)$$

Clearly, the quantity in each angular bracket is Hermitian, their ensemble average are thus real. Since $\hat{\rho}_0$ commute with $\hat{U}_0(t)$, the zeroth order term is

$$\mathbf{P}_0 = \langle \mu \rangle \quad (36)$$

This is just the permanent dipole of the molecule. Higher-order terms $\mathbf{P}^{(n)}(t)$ thus correspond to induced dipoles.

In nonlinear response theory, we will describe the electric field classically. The time-dependent perturbation is thus

$$\hat{V}(t) = -\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t) \quad (37)$$

Using the Einstein summation convention, this perturbation can be rewritten as

$$\hat{V}(t) = -\hat{\mu}_\xi \mathcal{E}_\xi(t) \quad (38)$$

We use ξ, η, ζ to label the Cartesian components of vectors and tensors. To avoid confusion with energy, we used $\mathcal{E}_\xi$ to denote the components of the electric field. The first-order induced dipole is

$$\begin{aligned} P_\xi^{(1)} &= \left\langle \hat{U}_0^\dagger(t) \hat{\mu}_\xi \hat{U}^{(1)}(t) + \hat{U}^{(1)\dagger}(t) \hat{\mu}_\xi \hat{U}_0(t) \right\rangle \\ &= \frac{i}{\hbar} \left\langle \int_{t_0}^t d\tau \hat{U}_0^\dagger(t) \hat{\mu}_\xi \hat{U}_0(t) \hat{U}_0^\dagger(\tau) \hat{\mu}_\eta \hat{U}_0(\tau) \mathcal{E}_\eta(\tau) \right\rangle \\ &\quad - \frac{i}{\hbar} \left\langle \int_{t_0}^t d\tau \hat{U}_0^\dagger(\tau) \hat{\mu}_\eta \hat{U}_0(\tau) \hat{U}_0^\dagger(t) \hat{\mu}_\xi \hat{U}_0(t) \mathcal{E}_\eta(\tau) \right\rangle \\ &= \int_{t_0}^t d\tau \left(\frac{i}{\hbar} \right) \langle [\hat{\mu}_\xi(t), \hat{\mu}_\eta(\tau)] \rangle \mathcal{E}_\eta(\tau) \\ &= \int_{t_0}^\infty d\tau \left(\frac{i}{\hbar} \right) \theta(t - \tau) \langle [\hat{\mu}_\xi(t - \tau), \hat{\mu}_\eta(0)] \rangle \mathcal{E}_\eta(\tau) \end{aligned} \quad (39)$$

where we have defined

$$\hat{\mu}_\xi(t) = \hat{U}_0^\dagger(t) \hat{\mu}_\xi \hat{U}_0(t) \quad (40)$$

In the last line, $\langle [\hat{\mu}_\xi(t), \hat{\mu}_\eta(\tau)] \rangle = \langle [\hat{\mu}_\xi(t - \tau), \hat{\mu}_\eta(0)] \rangle$ because $\hat{\rho}_0$ commute with $\hat{U}_0(\tau)$ and the trace is invariant under circular shifts. Taking $t_0 \rightarrow -\infty$ limit, we have

$$P_\xi^{(1)} = \int_{-\infty}^\infty R_{\xi\eta}^{(1)}(t - \tau) \mathcal{E}_\eta(\tau) d\tau \quad (41)$$

where $R_{\xi\eta}^{(1)}(t - \tau)$ is the first-order response function or **polarizability** of the molecule,

$$\boxed{R_{\xi\eta}^{(1)}(t - \tau) = \frac{i}{\hbar} \theta(t - \tau) \langle [\hat{\mu}_\xi(t - \tau), \hat{\mu}_\eta(0)] \rangle} \quad (42)$$

The second-order induced dipole is

$$\begin{aligned}
P_\xi^{(2)} &= \left\langle \hat{U}^{(1)\dagger}(t) \hat{\mu}_\xi \hat{U}^{(1)}(t) + \hat{U}_0^\dagger(t) \hat{\mu}_\xi \hat{U}^{(2)}(t) + \hat{U}^{(2)\dagger}(t) \hat{\mu}_\xi \hat{U}_0(t) \right\rangle \\
&= \frac{1}{\hbar^2} \int_{t_0}^t d\tau_2 \int_{t_0}^t d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&\quad - \frac{1}{\hbar^2} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \langle \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \hat{\mu}_\zeta(\tau_1) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&\quad - \frac{1}{\hbar^2} \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\eta(\tau_2) \hat{\mu}_\xi(t) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1)
\end{aligned} \tag{43}$$

Note that the first double integral has a different upper limit for τ_1 than the other two. This integral can be rewritten as

$$\begin{aligned}
&\int_{t_0}^t d\tau_2 \int_{t_0}^t d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&= \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&\quad + \int_{t_0}^t d\tau_2 \int_{\tau_2}^t d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&= \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \langle \hat{\mu}_\zeta(\tau_1) \hat{\mu}_\xi(t) \hat{\mu}_\eta(\tau_2) \rangle \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) \\
&\quad + \int_{t_0}^t d\tau_1 \int_{\tau_1}^t d\tau_2 \langle \hat{\mu}_\eta(\tau_2) \hat{\mu}_\xi(t) \hat{\mu}_\zeta(\tau_1) \rangle \mathcal{E}_\zeta(\tau_1) \mathcal{E}_\eta(\tau_2)
\end{aligned} \tag{44}$$

The last equality comes from interchanging integration variables and summation indices, $\tau_1 \leftrightarrow \tau_2$, $\eta \leftrightarrow \zeta$. The integration $\int_{t_0}^t d\tau_1 \int_{\tau_1}^t d\tau_2$ runs over the area where $t_0 < \tau_1 < \tau_2 < t$, which is the same as the integration $\int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1$. Therefore

$$\int_{t_0}^t d\tau_1 \int_{\tau_1}^t d\tau_2 \dots = \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \dots \tag{45}$$

Again, taking $t_0 \rightarrow -\infty$ limit, we have

$$P_\xi^{(2)} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{\xi\eta\zeta}^{(2)}(t - \tau_2, \tau_2 - \tau_1) \mathcal{E}_\eta(\tau_2) \mathcal{E}_\zeta(\tau_1) d\tau_1 d\tau_2 \tag{46}$$

where $R_{\xi\eta\zeta}^{(2)}(t - \tau_2, \tau_2 - \tau_1)$ is the second-order response function,

$$\boxed{R_{\xi\eta\zeta}^{(2)}(t - \tau_2, \tau_2 - \tau_1) = -\frac{1}{\hbar^2} \theta(t - \tau_2) \theta(\tau_2 - \tau_1) \langle [\hat{\mu}_\xi(t - \tau_1), \hat{\mu}_\eta(\tau_2 - \tau_1)], \hat{\mu}_\zeta(0) \rangle} \tag{47}$$

The expression of the third-order induced dipole contains integrals like

$$\int_{t_0}^t d\tau_3 \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \dots \tag{48}$$

The integration is over the volume defined by $t_0 < \tau_1 < \tau_2 < t$, $t_0 < \tau_3 < t$, it can be decomposed into three integrals with the time variables being fully ordered: $t_0 < \tau_1 < \tau_2 < \tau_3 < t$, $t_0 < \tau_3 < \tau_1 < \tau_2 < t$, and $t_0 < \tau_1 < \tau_3 < \tau_2 < t$. Therefore,

$$\begin{aligned} & \int_{t_0}^t d\tau_3 \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \dots \\ = & \int_{t_0}^t d\tau_3 \int_{t_0}^{\tau_3} d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \dots + \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \int_{t_0}^{\tau_1} d\tau_3 \dots + \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_3 \int_{t_0}^{\tau_3} d\tau_1 \dots \end{aligned} \quad (49)$$

Similar to the derivation for the second-order case, by interchanging integration variables and summation indices, we can obtain a compact form of the third-order response function. In general, the **n -th order response function** is

$$\begin{aligned} \mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1) &= \left(\frac{i}{\hbar}\right)^n \theta(t_n) \theta(t_{n-1}) \dots \theta(t_1) \\ &\quad \times \langle [\dots [\hat{\boldsymbol{\mu}}(t_n + \dots + t_1), \hat{\boldsymbol{\mu}}(t_{n-1} + \dots + t_1)] \dots], \hat{\boldsymbol{\mu}}(t_1)], \hat{\boldsymbol{\mu}}(0)] \rangle \end{aligned} \quad (50)$$

Here we introduced **time delays** as the variables,

$$t_{n-1} = \tau_n - \tau_{n-1}, \quad n = 2, 3, \dots; \quad t_n = t - \tau_n \quad (51)$$

In the commutator, the products of two dipole operators are tensor products. The n -th order response function $\mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1)$ is a $n + 1$ -dimensional tensor. **The commutators also lead to 2^n terms in the n -th order response function.**

Since $\mathcal{R}^{(n)}$ is real, these 2^n terms always appear in pair with their complex conjugates. In other words, **there are only 2^{n-1} independent terms in the n -th order response function.** For example, the second-order response function is

$$\begin{aligned} \mathcal{R}^{(2)}(t_2, t_1) &= -\frac{1}{\hbar^2} \theta(t_2) \theta(t_1) (\langle \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(0) \rangle - \langle \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(0) \rangle \\ &\quad + \langle \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(t_2 + t_1) \rangle - \langle \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(t_1) \rangle) \end{aligned} \quad (52)$$

Since both $\hat{\boldsymbol{\mu}}(t)$ and $\hat{\rho}_0$ are Hermitian and the trace is invariant under circular shifts, its easy to show that

$$C_1(t_2, t_1) \equiv \langle \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(0) \rangle = \langle \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(t_2 + t_1) \rangle^* \quad (53a)$$

$$C_2(t_2, t_1) \equiv -\langle \hat{\boldsymbol{\mu}}(t_1) \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(0) \rangle = -\langle \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t_2 + t_1) \hat{\boldsymbol{\mu}}(t_1) \rangle^* \quad (53b)$$

where C_1 and C_2 are correlation functions of the dipole moment. C_1 and C_2 are also tensors. $\mathcal{R}^{(2)}(t_2, t_1)$ can be written in terms of the correlation functions as

$$\mathcal{R}^{(2)}(t_2, t_1) = -\frac{1}{\hbar^2} \theta(t_2) \theta(t_1) \sum_{j=1}^2 (C_j(t_2, t_1) + C_j^*(t_2, t_1)) \quad (54)$$

In general, we have

$$\mathcal{R}^{(n)}(t_n, \dots, t_1) = \left(\frac{i}{\hbar}\right)^n \theta(t_n) \theta(t_{n-1}) \dots \theta(t_1) \sum_j^{2^{n-1}} (C_j(t_n, \dots, t_1) + (-1)^n C_j^*(t_n, \dots, t_1)) \quad (55)$$

4 Multiwave Mixing

4.1 Nonlinear Response Theory

The nonlinear response functions in frequency domain are given by Fourier transformation of $\mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1)$,

$$\mathcal{R}^{(n)}(\omega_n + \dots + \omega_1, \dots, \omega_2 + \omega_1, \omega_1) = \int dt_1 \int dt_2 \dots \int dt_n \mathcal{R}^{(n)}(t_n, t_{n-1}, \dots, t_1) \times e^{i(\omega_n + \dots + \omega_1)t_n} \times \dots \times e^{i(\omega_2 + \omega_1)t_2} \times e^{i\omega_1 t_1} \quad (56)$$

For simplicity, we will focus on the second-order response function from hereon,

$$\mathcal{R}^{(2)}(\omega_2 + \omega_1, \omega_1) = \int dt_1 \int dt_2 \mathcal{R}^{(2)}(t_2, t_1) e^{i(\omega_2 + \omega_1)t_2} e^{i\omega_1 t_1} \quad (57)$$

which can be written in terms of the Fourier transformed correlation functions,

$$\mathcal{R}^{(2)}(\omega_2 + \omega_1, \omega_1) = -\frac{1}{\hbar^2} \sum_{j=1}^2 (C_j(\omega_2 + \omega_1, \omega_1) + C_j^*(-\omega_2 - \omega_1, -\omega_1)) \quad (58)$$

where¹

$$C_j(\omega_2 + \omega_1, \omega_1) = \int dt_1 \int dt_2 \theta(t_2) \theta(t_1) C_j(t_2, t_1) e^{i(\omega_2 + \omega_1)t_2} e^{i\omega_1 t_1} \quad (59)$$

$$C_j^*(-\omega_2 - \omega_1, -\omega_1) = \int dt_1 \int dt_2 \theta(t_2) \theta(t_1) C_j^*(t_2, t_1) e^{i(\omega_2 + \omega_1)t_2} e^{i\omega_1 t_1} \quad (60)$$

Note that $C_j^*(-\omega_2 - \omega_1, -\omega_1)$ is not the complex conjugate of $C_j(\omega_2 + \omega_1, \omega_1)$ because the variables change signs. Therefore, $\mathcal{R}^{(2)}(\omega_2 + \omega_1, \omega_1)$ is not necessarily real. One can easily show that

$$\mathcal{R}^{(2)*}(\omega_2 + \omega_1, \omega_1) = \mathcal{R}^{(2)}(-\omega_2 - \omega_1, -\omega_1) \quad (61)$$

This relation holds for any n .

Now considering a molecule interacting with two plane waves with frequencies ω_1 and ω_2 , the electric field (omitting the $e^{i\mathbf{k}\cdot\mathbf{r}}$ part) is

$$\mathbf{E}(t) = \frac{1}{2} \mathbf{E}_1 (e^{i\omega_1 t} + e^{-i\omega_1 t}) + \frac{1}{2} \mathbf{E}_2 (e^{i\omega_2 t} + e^{-i\omega_2 t}) \quad (62)$$

The second-order induced dipole is

$$P_\xi^{(2)}(t) = \int dt_1 \int dt_2 R_{\xi\eta\zeta}^{(2)}(t_2, t_1) \mathcal{E}_\eta(t - t_2) \mathcal{E}_\zeta(t - t_2 - t_1) \quad (63)$$

Note that we switched back to the notation using components, because it is important to track the permutation of these components in subsequent derivations. Substituting the expression of $\mathbf{E}(t)$ and only keep the cross terms (i.e., the terms containing both $\mathbf{E}_1$ and $\mathbf{E}_2$), we have

¹In Mukamel's book, the Fourier transformation of the correlation functions includes an additional coefficient $(-i)^2$. However, despite a different definition used here, the resulted expression for the response function in frequency domain will be the same.

$$\begin{aligned}
P_\xi^{(2)}(t) &= \frac{1}{4} \left(R_{\xi\eta\zeta}^{(2)}(\omega_2 + \omega_1, \omega_1) e^{-i(\omega_1 + \omega_2)t} + R_{\xi\eta\zeta}^{(2)}(\omega_1 - \omega_2, \omega_1) e^{-i(\omega_1 - \omega_2)t} \right. \\
&\quad + R_{\xi\eta\zeta}^{(2)}(-\omega_2 - \omega_1, -\omega_1) e^{i(\omega_1 + \omega_2)t} + R_{\xi\eta\zeta}^{(2)}(-\omega_1 + \omega_2, -\omega_1) e^{i(\omega_1 - \omega_2)t} \\
&\quad + R_{\xi\zeta\eta}^{(2)}(\omega_2 + \omega_1, \omega_2) e^{-i(\omega_1 + \omega_2)t} + R_{\xi\zeta\eta}^{(2)}(\omega_2 - \omega_1, \omega_2) e^{-i(\omega_2 - \omega_1)t} \\
&\quad \left. + R_{\xi\zeta\eta}^{(2)}(-\omega_2 - \omega_1, -\omega_2) e^{i(\omega_1 + \omega_2)t} + R_{\xi\zeta\eta}^{(2)}(-\omega_2 + \omega_1, -\omega_2) e^{i(\omega_2 - \omega_1)t} \right) \mathcal{E}_1^\zeta \mathcal{E}_2^\eta \\
&= \frac{1}{2} \left(\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) e^{-i(\omega_1 + \omega_2)t} + \chi_{\xi\eta\zeta}^{(2)}(\omega_1, -\omega_2) e^{-i(\omega_1 - \omega_2)t} \right) \mathcal{E}_1^\zeta \mathcal{E}_2^\eta + \text{c.c.} \quad (64)
\end{aligned}$$

where

$$\boxed{\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) = \frac{1}{2} \left(R_{\xi\eta\zeta}^{(2)}(\omega_2 + \omega_1, \omega_1) + R_{\xi\zeta\eta}^{(2)}(\omega_1 + \omega_2, \omega_2) \right)} \quad (65)$$

is the symmetrized **hyperpolarizability**. $\mathcal{R}^{(2)}(\omega_2 + \omega_1, \omega_1)$ is asymmetric with respect to permutation of ω_1 and ω_2 because of the time-ordering in $\mathcal{R}^{(2)}(t_2, t_1)$. However, **in frequency-domain experiments, all time orders contribute**. $\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2)$ is constructed by permutation of the ω 's. **Note that the indices for the Cartesian components of the tensor should also be permuted.** $\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2)$ can also be expressed in terms of the correlation functions,

$$\begin{aligned}
\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) &= -\frac{1}{2\hbar^2} \sum_{j=1}^2 \left[(C_j^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) + C_j^{\xi\eta\zeta*}(-\omega_2 - \omega_1, -\omega_1)) \right. \\
&\quad \left. + (C_j^{\xi\zeta\eta}(\omega_1 + \omega_2, \omega_2) + C_j^{\xi\zeta\eta*}(-\omega_1 - \omega_2, -\omega_2)) \right] \quad (66)
\end{aligned}$$

Since in Eq. (64) we only keep the terms that contain both $\mathbf{E}_1$ and $\mathbf{E}_2$, we eliminated the cases where the molecule interact with the same radiation field twice. The second-order induced dipole moment has two oscillation frequencies, $\omega_1 + \omega_2$ and $|\omega_1 - \omega_2|$, which are the sum and difference of the frequencies of two incident light. These processes are known as the **sum frequency generation** and **difference frequency generation**. The second time derivative of $\mathbf{P}^{(2)}(t)$ is

$$\begin{aligned}
\ddot{P}_\xi^{(2)}(t) &= -\frac{1}{2} \left((\omega_1 + \omega_2)^2 \chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) e^{-i(\omega_1 + \omega_2)t} + (\omega_1 - \omega_2)^2 \chi_{\xi\eta\zeta}^{(2)}(\omega_1, -\omega_2) e^{-i(\omega_1 - \omega_2)t} \right) \mathcal{E}_1^\zeta \mathcal{E}_2^\eta \\
&\quad + \text{c.c.} \quad (67)
\end{aligned}$$

The signal intensity of sum frequency generation, for example, is thus

$$\boxed{I_{\text{SFG}} \propto \left| \chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) e_1^\zeta e_2^\eta \right|^2 I_1 I_2} \quad (68)$$

where I_1 and I_2 are intensities of the incident waves. $\mathbf{e}_1$ and $\mathbf{e}_2$ are unit vectors along the directions of $\mathbf{E}_1$ and $\mathbf{E}_2$. The SFG signal is proportional to the square of the second-order hyperpolarizability.

Next we explicitly calculate $\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2) e_1^\zeta e_2^\eta$. The density matrix in eigenstate representation is

$$\hat{\rho}_0 = \sum_a P_a |a\rangle \langle a| \quad (69)$$

where P_a is the Boltzmann population of state $|a\rangle$, the correlation functions can then be calculated using

$$\begin{aligned} C_1^{\xi\eta\zeta}(t_2, t_1) &= \sum_a P_a \langle a | \hat{\mu}_\xi(t_2 + t_1) \hat{\mu}_\eta(t_1) \hat{\mu}_\zeta(0) | a \rangle \\ &= \sum_{a,b,c} P_a \mu_{ac}^\xi \mu_{cb}^\eta \mu_{ba}^\zeta e^{i\omega_{ca}t_2} e^{i\omega_{ba}t_1} \end{aligned} \quad (70a)$$

$$\begin{aligned} C_2^{\xi\eta\zeta}(t_2, t_1) &= - \sum_a P_a \langle a | \hat{\mu}_\eta(t_1) \hat{\mu}_\xi(t_2 + t_1) \hat{\mu}_\zeta(0) | a \rangle \\ &= - \sum_{a,b,c} P_a \mu_{ac}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta e^{i\omega_{bc}t_2} e^{i\omega_{ba}t_1} \end{aligned} \quad (70b)$$

where $\omega_{ab} = \frac{E_b - E_a}{\hbar}$. The superscripts ξ , η and ζ denote the Cartesian components. Performing the Fourier transformation (first write the step function in its integral representation and then perform the integration over t), we have

$$C_1^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) = - \sum_{a,b,c} P_a \frac{\mu_{ac}^\xi \mu_{cb}^\eta \mu_{ba}^\zeta}{(\omega_1 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ca} + i\varepsilon)} \quad (71a)$$

$$C_2^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) = \sum_{a,b,c} P_a \frac{\mu_{ac}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta}{(\omega_1 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{bc} + i\varepsilon)} \quad (71b)$$

The other terms can be directly obtained from these two expressions using the relations discussed above. Note that sometimes we need to interchange the dummy indices b and c .

$$C_1^{\xi\eta\zeta*}(-\omega_2 - \omega_1, -\omega_1) = - \sum_{a,b,c} P_a \frac{\mu_{ac}^\zeta \mu_{cb}^\eta \mu_{ba}^\xi}{(\omega_1 + \omega_{ac} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ab} + i\varepsilon)} \quad (72a)$$

$$C_2^{\xi\eta\zeta*}(-\omega_2 - \omega_1, -\omega_1) = \sum_{a,b,c} P_a \frac{\mu_{ac}^\zeta \mu_{cb}^\xi \mu_{ba}^\eta}{(\omega_1 + \omega_{ac} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{bc} + i\varepsilon)} \quad (72b)$$

$$C_1^{\xi\zeta\eta}(\omega_2 + \omega_1, \omega_2) = - \sum_{a,b,c} P_a \frac{\mu_{ac}^\xi \mu_{cb}^\zeta \mu_{ba}^\eta}{(\omega_2 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ca} + i\varepsilon)} \quad (72c)$$

$$C_2^{\xi\zeta\eta}(\omega_2 + \omega_1, \omega_2) = \sum_{a,b,c} P_a \frac{\mu_{ac}^\zeta \mu_{cb}^\xi \mu_{ba}^\eta}{(\omega_2 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{bc} + i\varepsilon)} \quad (72d)$$

$$C_1^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_2) = - \sum_{a,b,c} P_a \frac{\mu_{ac}^\eta \mu_{cb}^\zeta \mu_{ba}^\xi}{(\omega_2 + \omega_{ac} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ab} + i\varepsilon)} \quad (72e)$$

$$C_2^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_2) = \sum_{a,b,c} P_a \frac{\mu_{ac}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta}{(\omega_2 + \omega_{ac} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{bc} + i\varepsilon)} \quad (72f)$$

and $\chi_{\xi\eta\zeta}^{(2)}(\omega_1, \omega_2)$ equals to $-\frac{1}{2\hbar^2}$ times the sum of all 8 correlation functions.

4.2 Time-dependent Perturbation Theory

In time-dependent perturbation theory, SFG refers to the transition

$$|a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |d\mathbf{k}_s\rangle \quad (73)$$

where $\mathbf{k}_1$ and $\mathbf{k}_2$ are wave vectors of the two incident lights, and $\mathbf{k}_s$ is the wave vector of the signal. We assume that the photon occupation numbers equal to 1 for all three wave vectors. Since three occupation numbers change during SFG, a third-order approximation is needed, the third order coupling for this process is

$$V_{a \rightarrow d}^{(3)} = \sum_{b,c} \frac{V_{dc}V_{cb}V_{ba}}{(E_a - E_c + i\varepsilon)(E_a - E_b + i\varepsilon)} \quad (74)$$

where b and c denotes intermediate molecule-photon states. It's easy to show that only six pathways contribute to $V_{a \rightarrow d}^{(3)}$:

$$\begin{aligned} (1) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_2\rangle \rightarrow |c\rangle \rightarrow |d\mathbf{k}_s\rangle \\ (2) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_1\rangle \rightarrow |c\rangle \rightarrow |d\mathbf{k}_s\rangle \\ (3) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_1\mathbf{k}_2\mathbf{k}_s\rangle \rightarrow |c\mathbf{k}_2\mathbf{k}_s\rangle \rightarrow |d\mathbf{k}_s\rangle \\ (4) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_1\mathbf{k}_2\mathbf{k}_s\rangle \rightarrow |c\mathbf{k}_1\mathbf{k}_s\rangle \rightarrow |d\mathbf{k}_s\rangle \\ (5) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_2\rangle \rightarrow |c\mathbf{k}_2\mathbf{k}_s\rangle \rightarrow |d\mathbf{k}_s\rangle \\ (6) \quad & |a\mathbf{k}_1\mathbf{k}_2\rangle \rightarrow |b\mathbf{k}_1\rangle \rightarrow |c\mathbf{k}_1\mathbf{k}_s\rangle \rightarrow |d\mathbf{k}_s\rangle \end{aligned} \quad (75)$$

We can use the properties of photon creation and annihilation operators to calculate these coupling terms, the results are

$$V_1^{\xi\eta\zeta} = i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\xi \mu_{cb}^\eta \mu_{ba}^\zeta}{(\omega_1 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ca} + i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76a)$$

$$V_2^{\xi\eta\zeta} = i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\xi \mu_{cb}^\zeta \mu_{ba}^\eta}{(\omega_2 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_2 + \omega_{ca} + i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76b)$$

$$V_3^{\xi\eta\zeta} = i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\eta \mu_{cb}^\zeta \mu_{ba}^\xi}{(\omega_2 + \omega_{dc} - i\varepsilon)(\omega_1 + \omega_2 + \omega_{db} - i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76c)$$

$$V_4^{\xi\eta\zeta} = i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\zeta \mu_{cb}^\eta \mu_{ba}^\xi}{(\omega_1 + \omega_{dc} - i\varepsilon)(\omega_1 + \omega_2 + \omega_{db} - i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76d)$$

$$V_5^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta}{(\omega_1 + \omega_{ba} + i\varepsilon)(\omega_2 + \omega_{dc} - i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76e)$$

$$V_6^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \sum_{b,c} \frac{\mu_{dc}^\zeta \mu_{cb}^\xi \mu_{ba}^\eta}{(\omega_2 + \omega_{ba} + i\varepsilon)(\omega_1 + \omega_{dc} - i\varepsilon)} e_1^\zeta e_2^\eta e_s^\xi \quad (76f)$$

In these equations, we have used the relation

$$\omega_s = \omega_1 + \omega_2 + \omega_{da} \quad (77)$$

which is ensured by the delta function in the rate constant expression. Obviously, if we set $|d\rangle = |a\rangle$ in the time-dependent perturbation theory and set $\hat{\rho}_0 = |a\rangle\langle a|$ (a pure state) in the nonlinear response theory, taking $\varepsilon \rightarrow 0$ limit, we have

$$V_1^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} C_1^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) e_1^\zeta e_2^\eta e_s^\xi \quad (78a)$$

$$V_2^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} C_1^{\xi\zeta\eta}(\omega_2 + \omega_1, \omega_2) e_1^\zeta e_2^\eta e_s^\xi \quad (78b)$$

$$V_3^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} C_1^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_2) e_1^\zeta e_2^\eta e_s^\xi \quad (78c)$$

$$V_4^{\xi\eta\zeta} = -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} C_1^{\xi\eta\zeta*}(-\omega_2 - \omega_1, -\omega_1) e_1^\zeta e_2^\eta e_s^\xi \quad (78d)$$

On the other hand, when $\hat{\rho}_0 = |a\rangle\langle a|$, $\varepsilon \rightarrow 0$,

$$\begin{aligned} C_2^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) + C_2^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_2) &= \sum_{b,c} \frac{\mu_{ac}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta}{\omega_1 + \omega_2 + \omega_{bc}} \left(\frac{1}{\omega_2 + \omega_{ac}} + \frac{1}{\omega_1 + \omega_{ba}} \right) \\ &= \sum_{b,c} \frac{\mu_{ac}^\eta \mu_{cb}^\xi \mu_{ba}^\zeta}{(\omega_1 + \omega_{ba})(\omega_2 + \omega_{ac})} \end{aligned} \quad (79a)$$

$$\begin{aligned} C_2^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_2) + C_2^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_1) &= \sum_{b,c} \frac{\mu_{ac}^\zeta \mu_{cb}^\xi \mu_{ba}^\eta}{\omega_1 + \omega_2 + \omega_{bc}} \left(\frac{1}{\omega_1 + \omega_{ac}} + \frac{1}{\omega_2 + \omega_{ba}} \right) \\ &= \sum_{b,c} \frac{\mu_{ac}^\zeta \mu_{cb}^\xi \mu_{ba}^\eta}{(\omega_2 + \omega_{ba})(\omega_1 + \omega_{ac})} \end{aligned} \quad (79b)$$

Therefore, when $|d\rangle = |a\rangle$ in the time-dependent perturbation theory,

$$\begin{aligned} V_5^{\xi\eta\zeta} &= -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \\ &\quad \times \left(C_2^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_1) + C_2^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_2) \right) e_1^\zeta e_2^\eta e_s^\xi \end{aligned} \quad (80a)$$

$$\begin{aligned} V_6^{\xi\eta\zeta} &= -i\sqrt{\omega_1\omega_2\omega_s} \left(\frac{\hbar}{2Vn_r^2\varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \\ &\quad \times \left(C_2^{\xi\eta\zeta}(\omega_2 + \omega_1, \omega_2) + C_2^{\xi\zeta\eta*}(-\omega_2 - \omega_1, -\omega_1) \right) e_1^\zeta e_2^\eta e_s^\xi \end{aligned} \quad (80b)$$

Clearly, $V^{(3)}$ covers all eight correlation functions in $\chi^{(2)}(\omega_1, \omega_2)$. The C_2 correlation functions interfere with each other, leading to different looking terms in $V^{(3)}$. If we define an second-order hyperpolarizability operator, $\hat{\chi}^{(2)}(\omega_1, \omega_2)$, then

$$\chi^{(2)}(\omega_1, \omega_2) = \langle \hat{\chi}^{(2)}(\omega_1, \omega_2) \rangle = \text{Tr} \left(\hat{\chi}^{(2)}(\omega_1, \omega_2) \hat{\rho}_0 \right) \quad (81)$$

$$V_{a \rightarrow d}^{(3)} = -i\sqrt{\omega_1 \omega_2 \omega_s} \left(\frac{\hbar}{2V n_T^2 \varepsilon_0} \right)^{\frac{3}{2}} e^{i(\mathbf{k}_s - \mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} \langle d | \hat{\chi}_{\xi \eta \zeta}^{(2)}(\omega_1, \omega_2) | a \rangle e_1^\zeta e_2^\eta e_s^\xi \quad (82)$$

The third-order coupling term in nonlinear time-dependent response theory is proportional to the matrix element of the second-order hyperpolarizability. Both theories thus provide the same results for the SFG signal intensity. In general, for $n + 1$ -wave mixing process,

$$V_{a \rightarrow b}^{(n+1)} \propto \langle b | \hat{\chi}^{(n)}(\omega_1, \dots, \omega_n) | a \rangle \cdot \mathbf{e}_1 \cdot \dots \cdot \mathbf{e}_n \cdot \mathbf{e}_s \quad (83)$$

where $\hat{\chi}^{(n)}(\omega_1, \dots, \omega_n)$ is the symmetrized n th-order hyperpolarizability that encompasses all permutations of ω 's. Note that when permuting frequencies, the corresponding indices of the $\hat{\mathcal{R}}$ tensor also need to be permuted, *vide supra*.

Finally, we will discuss the advantages and limitations of the two approaches. In time-dependent perturbation theory, the signal frequency can be different from the sum/difference of the incident light frequencies, because the final state of the molecule after multiwave mixing does not need to be the same as its initial state. However, in nonlinear response theory, the signal frequency always equals to the combination of the sum/difference of the incident frequencies. Similar to Raman spectrum, in nonlinear response theory, shifts in signal frequency can be described as $\chi^{(n)}$ having an intrinsic oscillation frequency caused by molecular vibrations. In other words, in time-dependent perturbation theory, the influence of vibrational transitions to the signal is described quantum mechanically using “transition hyperpolarizability”, while in nonlinear response theory, the influence of vibrational transitions to the signal is treated classically. On the other hand, time-dependent perturbation theory approach has several limitations. First, it can only describe full frequency domain processes, thus provides no temporal information. Most importantly, **the transition rate is defined according to Pauli master equation. For a mixed state, it only describes the evolution of diagonal elements of the density matrix. The quantum coherence and dephasing of the process is completely missing.**